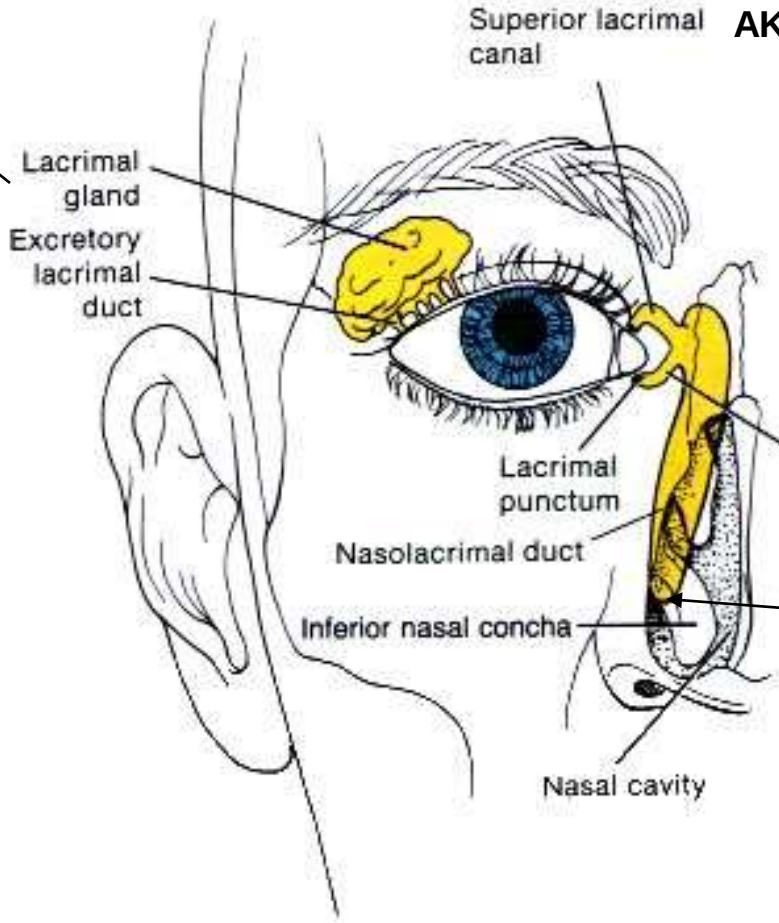
Lacrimal System Diseases

Lorne Yudcovitch, OD, MS, FAAO

Lacrimal System

Two lobes: palpebral and orbital



AKA Superior Canaliculus

AKA Inferior Canaliculus

Valve of Hasner

Congenital Dacryocele

- Collection of amniotic fluid within an obstructed lacrimal sac
- Appears as a firm bluish mass on the side of the nose and inferior to the medial canthus of the eyelid
- Hot compresses and massage is the initial treatment but most infants will require probing



Congenital Nasolacrimal Duct Obstruction

- A very common disorder in infants; occurs clinically in 2-4% full-term infants
- Results from a failure of the nasolacrimal duct to completely canalize during gestation
- Usually the obstruction is at the nasal end, the “valve” of Hasner
- Unilateral or bilateral
- Most common cause of tearing in childhood
- Can lead to a bacterial dacryocystitis and conjunctivitis

Congenital Nasolacrimal Duct Obstruction

- Tearing
- Discharge
- Crusting
- Mucus reflux from the punctum with compression over the lacrimal sac



Congenital Nasolacrimal Duct Obstruction

Management

- **Warm compresses bid to qid**
- **Topical antibiotic ointment (erythromycin bid for 1 week) if mucopurulent discharge**
- **Massage bid to qid - parent places index finger over infant's canaliculi (medial corner of eyelid) and makes several slow downward strokes**
  - **The motion will send hydrostatic forces down the nasolacrimal duct to open the obstruction**

Congenital Nasolacrimal Duct Obstruction

- Over 90% of nasolacrimal obstruction will resolve spontaneously or with the assistance of nasolacrimal duct massage
- Probing is considered between 6 and 13 months of age; exact timing for probing is controversial
- If probing is not successful, treatment is usually surgical

Probing Of A Congenital Nasolacrimal Duct Obstruction



Congenital Glaucoma

- **DDx congenital tearing**
- **If the child has both tearing and photophobia, the possibility of congenital glaucoma should be considered, even if the cornea is not cloudy or large**



Congenital Glaucoma



Acquired Nasolacrimal Duct Obstruction

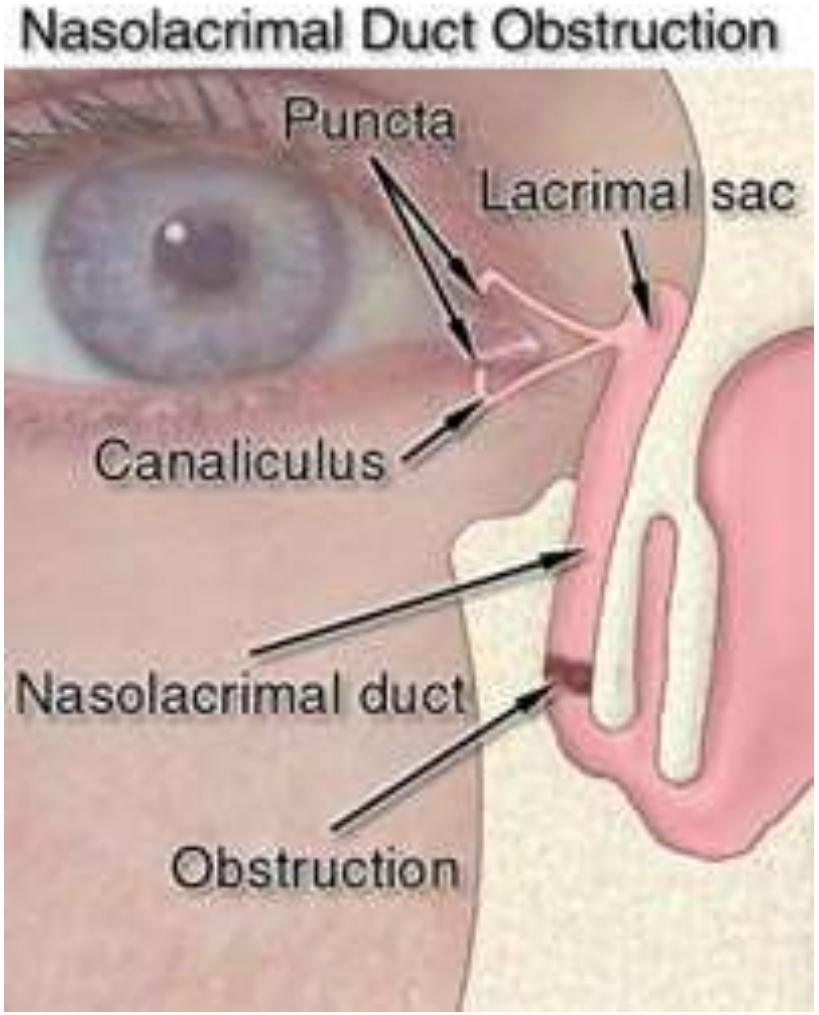
- **Etiology**
  - **Chronic sinus disease**
  - **Age-related stenosis**
  - **Naso/orbital trauma**
- **Findings**
  - **Tearing**
  - **Discharge**
  - **Crusting**
  - **Recurrent conjunctivitis**
- **Dx - Jones test(s)**
- **Often leads to acute dacryocystitis**

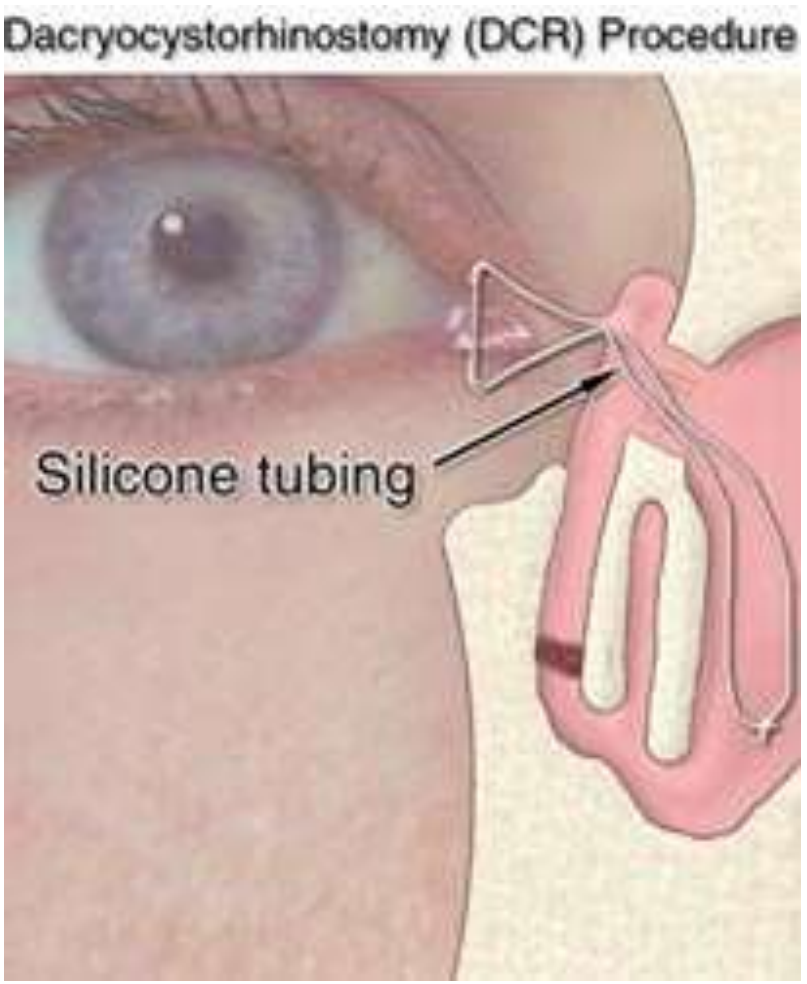
Acquired Nasolacrimal Duct Obstruction

Management

- **Warm compresses**
- **Topical antibiotic-steroid if partial obstruction**
- **Dilation/irrigation and/or probing**
- **If not successful, management is usually surgical**
  - **Silicone intubation**
  - **Dacryocystorhinostomy (anastomosis between lacrimal sac and nasal cavity)**

Dacryocystorhinostomy (DCR)





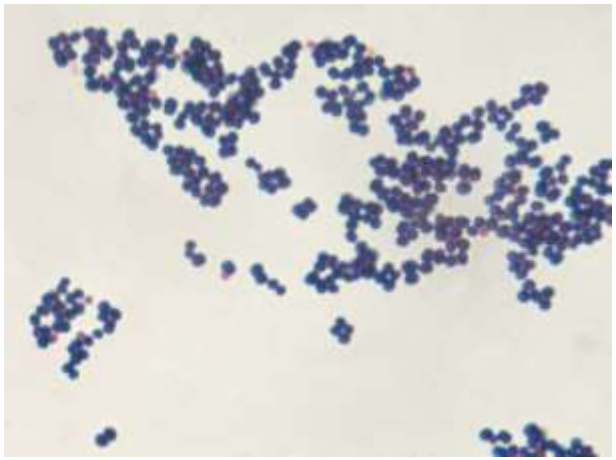
Acute Dacryocystitis

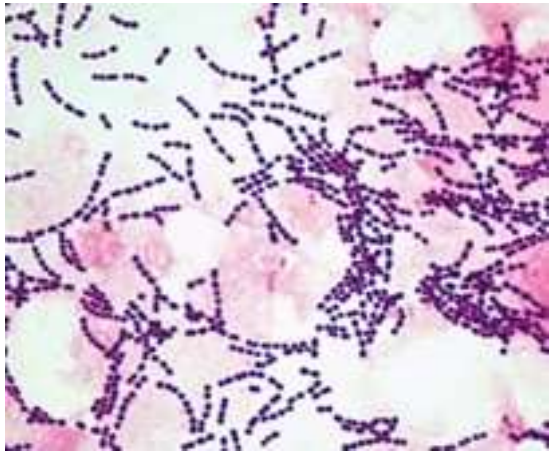
- Infection of the lacrimal sac
- Etiology
  - Nasolacrimal duct obstruction →
  - Tears stagnate in lacrimal sac
  - Become infected
- Causes of nasolacrimal duct obstruction
  - Long and narrow nasolacrimal ducts
  - Lacrimal sac diverticulum
  - Trauma
  - Dacryoliths
  - Inflammatory sinus and nasal problems

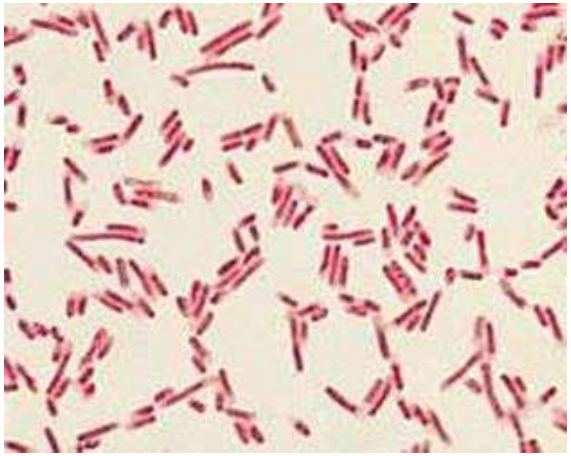
Acute Dacryocystitis

Bacteria causing infection

- **Staphylococci**
- **Streptococci**
- **E. coli, Pseudomonas, other Gram negatives**







Acute Dacryocystitis

Findings

- **Tenderness, erythema, and swelling over nasal portion of lower eyelid**
- **Epiphora**
- **Purulent punctal discharge particularly if pressure is applied to the lacrimal sac**
- **Fever and malaise may occur**

Complications

- **Preseptal cellulitis**
- **Conjunctivitis**
- **Cutaneous fistulas may form from the lacrimal sac**
- **Less common, orbital cellulitis and sepsis**

Acute Dacryocystitis

- **Tender, red, bump in the area of the lacrimal sac**



Acute Dacryocystitis





Acute Dacryocystitis



Acute Dacryocystitis





Acute Dacryocystitis





Acute Dacryocystitis

- Warm compress tid
- Systemic antibiotic
  - Oral Augmentin (amoxicillin/clavulante)
  - IV Unasyn (amoxicillin/sulbactam)
- Topical antibiotic ointment (erythromycin) if conjunctivitis exists
- Aspirate lacrimal sac contents with 19 gauge needle for culture and susceptibility testing

Acute Dacryocystitis

- If pointing abscess, consider incision and drainage
- After infection resolved, lacrimal system dilation/irrigation and/or probing to relieve the obstruction
- Nasal exam, possible CT scan
- Consider dacryocystorhinostomy (DCR) with silicon tubing to correct the nasolacrimal duct obstruction and prevent future dacryocystitis



Chronic Dacryocystitis

- Enlarged lacrimal sac without signs of acute inflammation
- Etiology
  - Chronic infection
  - Chronic inflammation/granuloma
  - Tumor (cancer)
- Culture to determine antibiotic therapy; then dacryocystorhinostomy
- If not infectious, biopsy and appropriate treatment

Chronic Dacryocystitis



Carcinoma Of The Lacrimal Sac



Acute Dacryoadenitis

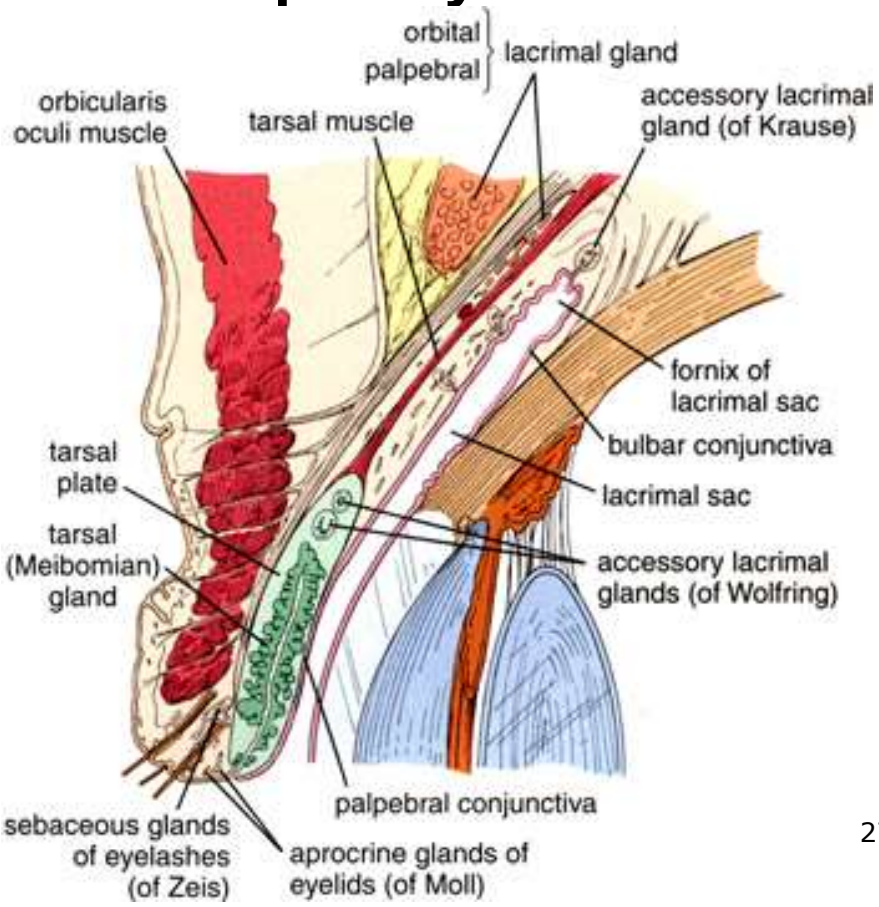
Inflammation of the lacrimal gland

- **Palpebral lobe affected more frequently than orbital**

Most cases associated with systemic infection

Etiology

- **Infection**
  - **Viral infection**
    - **Mumps**
    - **Epstein-Barr virus**
    - **Herpes zoster**
  - **Bacterial infection**
    - **Staphylococci**
    - ***Neisseria gonorrhoeae***



Acute Dacryoadenitis

Clinical Features

- **Temporal upper eyelid inflammation**
- **Tearing**
- **Discharge**
- **Fever**
- **Proptosis if orbital lobe involved**



Work up

- **Culture and Gram staining of discharge**
- **CBC**
- **Blood culture**
- **Consider orbital CT scan**

Acute Dacryoadenitis

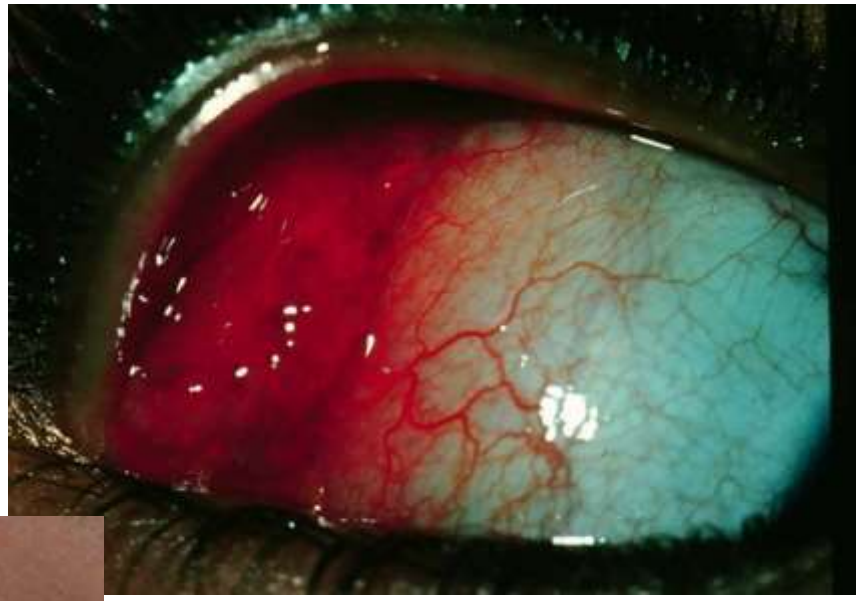
- **Upper lid**
  - **Swelling**
  - **Redness**
  - **Warmth**
  - **Tenderness to palpation**
  - **S-shaped deformity**





Acute Dacryoadenitis

- **Swollen palpebral lobe of the lacrimal gland can be easily visualized in the supratemporal fornix by elevating the upper lid while having the patient looking down**





Acute Dacryoadenitis

Management - Treat the underlying infection or inflammation

- **Mumps/Epstein-Barr virus - warm compresses**
- **Herpes simplex/zoster virus - systemic acyclovir or famciclovir**
- **Staphylococci and streptococci - systemic antibiotic eg Augmentin**
- ***Neisseria gonorrhoeae* - systemic antibiotic eg ceftriaxone**
- ***Mycobacterium* species - surgical excision + systemic antibiotics eg isoniazid and rifampin**
- ***Treponema pallidum* - systemic antibiotic eg penicillin G**

Chronic Dacryoadenitis

Chronic inflammation of the lacrimal gland

- **More common than acute**
- **Enlarged lacrimal gland**
- **Globe displacement**
- **Restricted ocular motility**

Etiology

- **Sarcoidosis**
- **Thyroid ophthalmopathy**
- **Sjogren's syndrome**
- **Tumors**

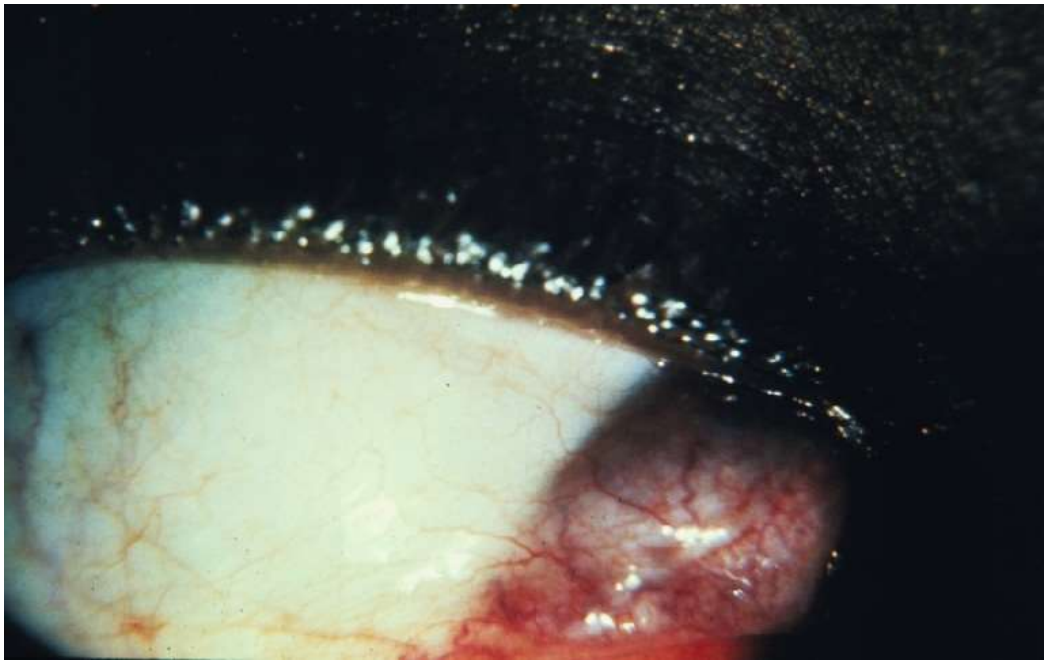


Bilateral Dacryoadenitis



Chronic Dacryoadenitis - Sarcoid

- **Lacrimal gland sarcoidosis (granulomatous inflammation)**



Sjogrens Syndrome

Autoimmune disease

Triad: dry eye, dry mouth, arthritis

Lacrimal gland infiltrated with lymphocytes → may be enlarged



Chronic Dacryoadenitis

Lacrimal gland carcinoma





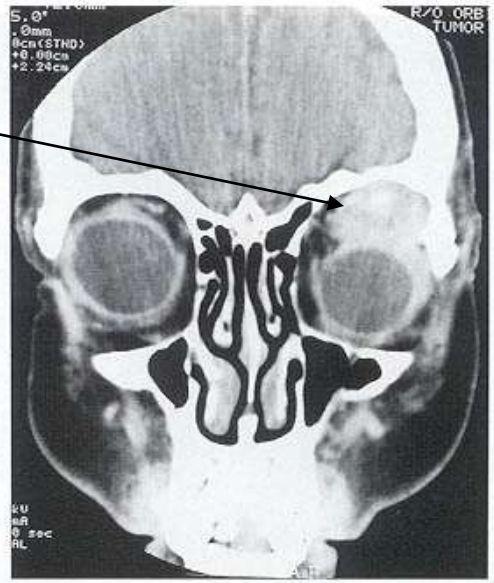
Lacrimal gland tumor

Lacrimal Gland Tumors

- **Enlargement of lacrimal fossa with displacement of the globe and no inflammatory signs**

CT scan

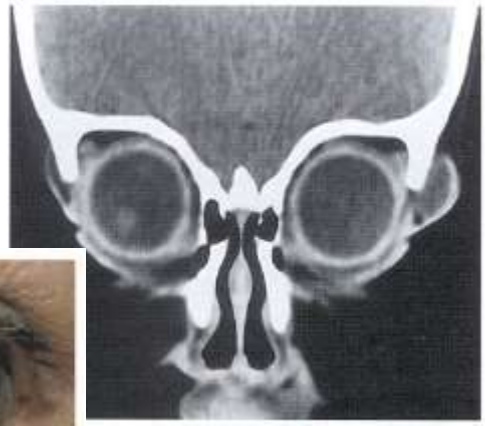


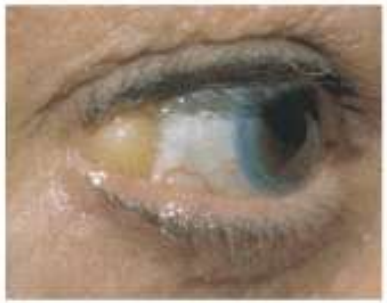


DDx of Chronic Dacryoadenitis - Dermoid Cyst

- Cystic lesion seen in preschool children
- Superotemporal mass
- Mobile and nontender
- Well circumscribed on CT with rare bony remodeling
- Surgical excision

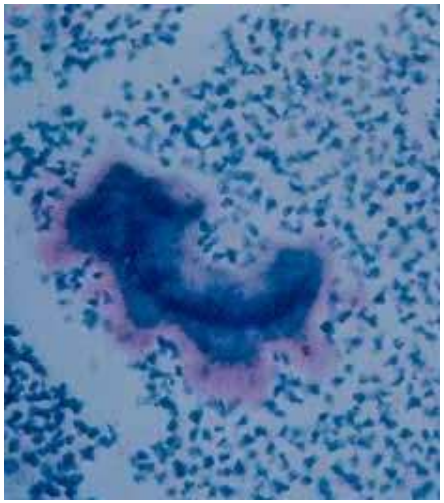






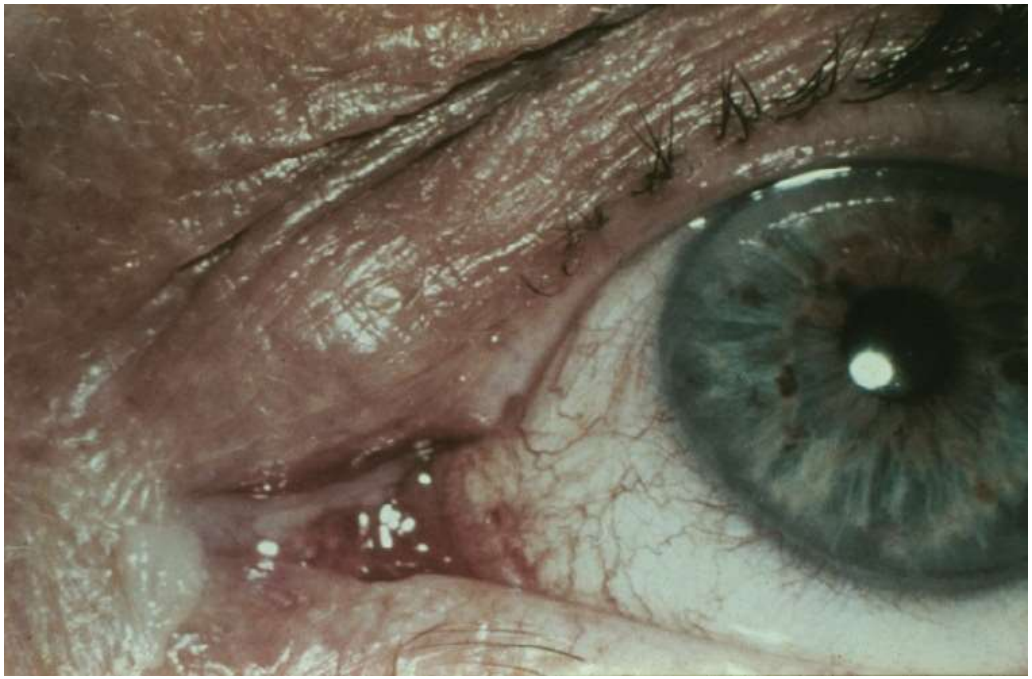
Canaliculitis

- **Infection of the canaliculus (duct between the punctum and the lacrimal sac)**
- **Etiology**
  - ***Actinomyces Israelii* (filamentous gram + rod)**
  - ***Candida albicans* (yeast)**
  - ***Aspergillus* (filamentous fungus)**
  - ***Nocardia asteroides (*filamentous rod)**
  - **Herpes simplex/zoster (viruses)**



Canaliculitis

- Erythema and swelling of punctum and adjacent tissues
- Conjunctivitis around the medial canthus



Canaliculitis



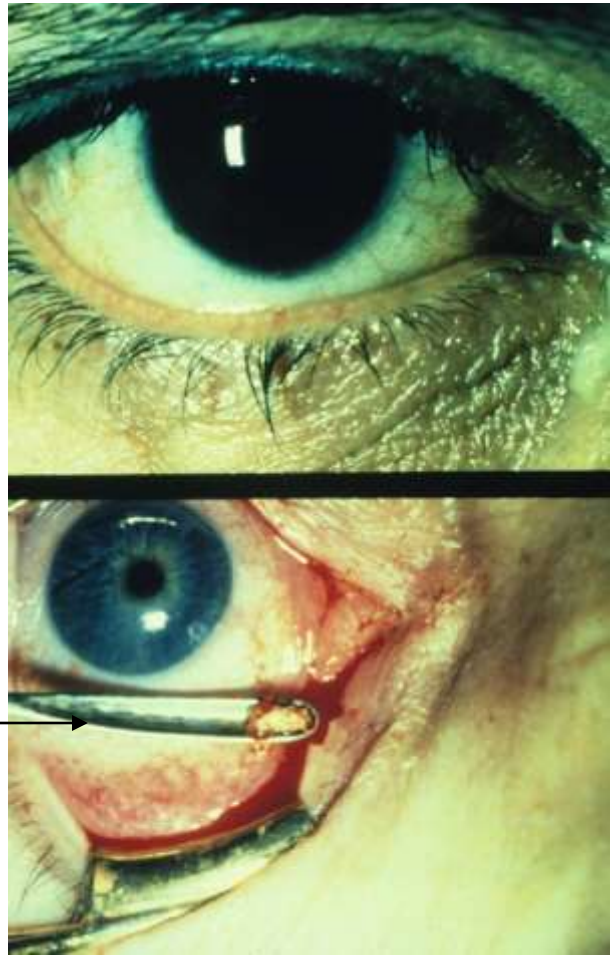


Expression of discharge from the punctum

Canaliculitis - Bottom Shows Removal Of Sulfur Granules

Canalicular concretion removal either by manual expression through the punctum or by scraping the canaliculus

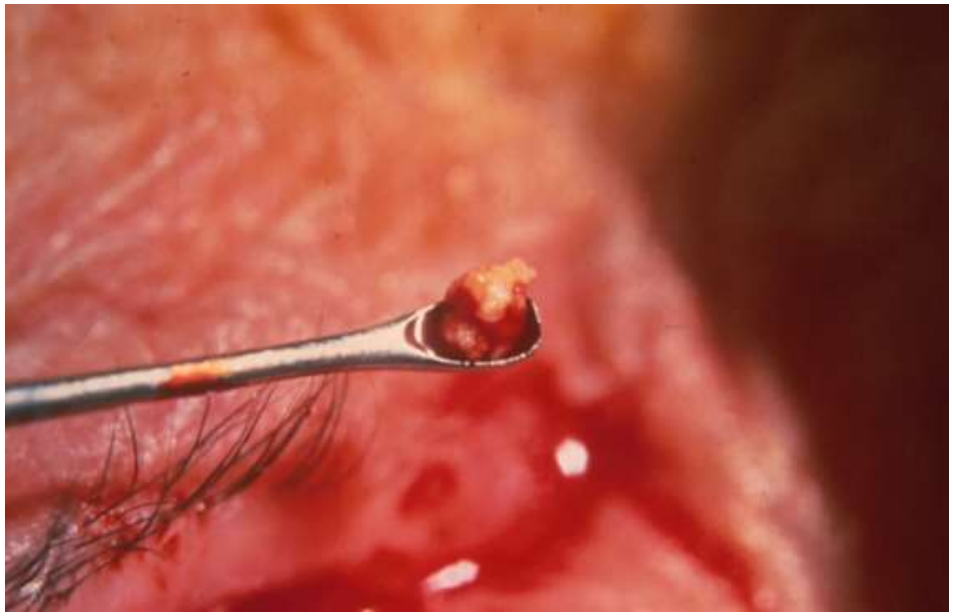
Here sulfur granules associated with *Actinomyces* infection are removed



Canaliculitis

Management

- **Warm compresses**
- **Culture**
- **Remove concretions**
- **Appropriate systemic and topical antibiotics**



Jones Test To Determine Patency Of Drainage System



THANK YOU!